

Monte Carlo Corrected Index Learning for Restless and Rested Bandits

Bias-Corrected Whittle and Gittins Index Learning

Your Name

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Problem Setting

Restless Multi-Armed Bandits (RMAB)

At every timestep:

- There are N arms.
- Only K arms can be activated (budget constraint).
- Each arm has state $s \in \mathcal{S}$.
- Goal: maximize long-term cumulative reward.

Challenge with existing methods:

- QWI/QGI use temporal-difference (TD) bootstrapping.
- Bootstrapped targets can be biased early in training.
- Wrong Q estimates $\rightarrow$ wrong index ranking.

Research Question

Can Monte Carlo returns reduce bias and improve index learning?

Standard TD Target:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

Problem:

- If $Q(s', a')$ is inaccurate, target becomes inaccurate.
- Errors propagate across updates.
- This causes:
 - slow convergence
 - unstable arm ranking
 - poor early exploration

Our Idea: Monte Carlo Corrected Q Learning

Instead of trusting only TD estimates, combine:

- Bootstrapped Q-values
- Monte Carlo returns

Monte Carlo Return:

$$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots$$

MC-Corrected Q-value:

$$\tilde{Q}_i(s, a) = \lambda_i(s, a) \hat{G}_i(s, a) + (1 - \lambda_i(s, a)) Q_i(s, a)$$

where:

$$\lambda_i(s, a) = \frac{n_i(s, a)}{n_i(s, a) + N_0}$$

QGI (Rested Bandits)

$$Q_i(s, a) \leftarrow Q_i(s, a) + \alpha [r + \gamma \max_{a'} Q_i(s', a') - Q_i(s, a)]$$

$$GI_i(s) = \frac{Q_i(s, 1) - Q_i(s, 0)}{1 - \gamma}$$

QWI (Restless Bandits)

$$Q_i(s, a) \leftarrow Q_i(s, a) + \alpha [r^{eff} + \gamma \max_{a'} Q_i(s', a') - Q_i(s, a)]$$

$$WI_i(s) = Q_i(s, 1) - Q_i(s, 0)$$

MC-QGI Update

$$Q_i(s, a) \leftarrow Q_i(s, a) + \alpha \left[r + \gamma \max_{a'} \tilde{Q}_i(s', a') - Q_i(s, a) \right]$$

$$MCQI_i(s) = \frac{\tilde{Q}_i(s, 1) - \tilde{Q}_i(s, 0)}{1 - \gamma}$$

MC-QWI Update

$$Q_i(s, a) \leftarrow Q_i(s, a) + \alpha \left[r^{eff} + \gamma \max_{a'} \tilde{Q}_i(s', a') - Q_i(s, a) \right]$$

$$MCWI_i(s) = \tilde{Q}_i(s, 1) - \tilde{Q}_i(s, 0)$$

Subsidy μ (QWI) and Retirement M (QGI)

QWI — subsidy

Diagonal MC-corrected gap (double-Q average at $s=x$):

$$\Delta_i(x) = \frac{1}{2} \left[\tilde{Q}_i^A(x, x, 1) - \tilde{Q}_i^A(x, x, 0) + \tilde{Q}_i^B(x, x, 1) - \tilde{Q}_i^B(x, x, 0) \right].$$

$$\mu_i(x) \leftarrow \text{clip}(\mu_i(x) + \beta \Delta_i(x)).$$

Base reward in the augmented update: $b_i(x, a) = (1 - a) \mu_i(x) + a r_i$.

QGI — retirement

For active value $Q_i^x(s, 1)$ at reference x , diagonal $s=x$:

$$M_i(x) \leftarrow \text{clip}(M_i(x) + \beta (Q_i^x(x, 1) - M_i(x))).$$

Algorithm Flow

For each episode:

① **Explore Arms**

Use behavior policy π_β .

② **Collect Returns**

Compute Monte Carlo returns:

$$G_t$$

③ **Update MC-Corrected Q**

Compute:

$$\tilde{Q}$$

Update Q-values.

④ **Compute Indices**

Compute:

Novel Contributions

- First MC-corrected index learning for QWI/QGI.
- Combines:
 - Fast TD convergence
 - Unbiased MC estimates
- Better arm ranking during early exploration.
- Applicable to:
 - Rested bandits
 - Restless bandits

Evaluation Metrics

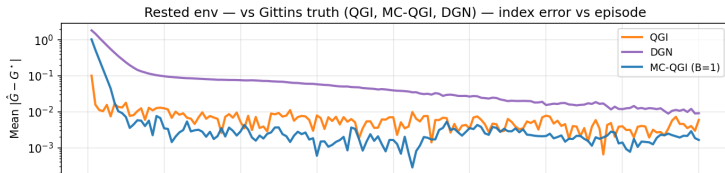
Compared against:

- QWI
- QGI
- QWINN
- DGN

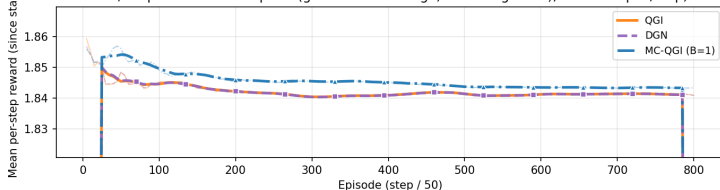
Measured:

- Final average reward
- Convergence speed
- Variance across seeds
- Compute per episode

Results: rested vs Gittins (index error & reward)

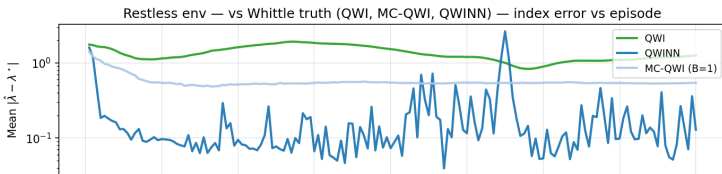


Reward: cumulative / steps at each checkpoint (global time-average; bold = light MA); tabular 1 pull/step; MC Σ top-f

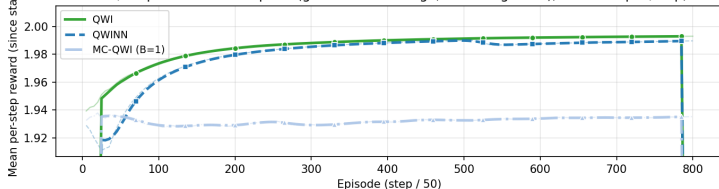


K=5 arms N=5 states/arm $\gamma=0.9$ steps=40000 log_every=250 seed=42 | mc_episode_len=50 smooth_w=1 track_reward=global_avg
MC: top-B concurrent activations per step, $B \in \{1\}$ (same K,N as tabular; tabular uses 1 arm/step).
env: RMABEnvironment.make_toy - rested track vs Gittins; restless track vs Whittle (same P,R, γ per K,N).

Results: restless vs Whittle (index error & reward)



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Key Insight

Using Monte Carlo returns to correct index learning reduces TD bias and improves arm ranking.

Future Work

- Hybrid with Policy Gradient
- Hybrid with LP-based policies
- Deep neural index approximators